

Mathematics Stage 4

Equations

Book 2 Solve and verify linear equations by substitution

Version: 261001

Feedback:
<https://MrDingMaths.com>

Contents

Syllabus Content.....	2
Substitution into Formulas.....	3
Solving for an Unknown after Substitution	8

Syllabus Content

MA4-EQU-C-01 solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$

Solve and verify linear equations by substitution

- Solve problems involving linear equations, including those arising from substituting given values into formulas

Substitution into Formulas

Review

☐ Substitute a value into an expression

a Substitute $x = 3$ into
 $y = 2x + 5$

b Substitute $x = 5, y = 7$ into
 $A = xy$

c Substitute $x = 15$ into
 $A = \frac{x}{3} + 2$

d Substitute $x = -5$ into
 $y = -2x + 1$

e Substitute $x = 15, y = -3$ into
 $C = \frac{x}{2-y}$

f Substitute $m = 3, n = 7$ into
 $R = \sqrt{6m + n}$

Formulas

A **formula** is a type of equation that relates two or more variables.



$$5 = 2x - 7$$



$$y = 2x - 7$$



$$A = 2\pi r^2 + 2\pi rh$$



$$x = 3$$

Two formulas you would be familiar with are:

Area of a rectangle: $A = lw$

Circumference of a circle: $C = 2\pi r$

The **subject** is the **isolated variable** on the **left-hand side**.

Key Ideas

1. An equation that shows a relationship between two or more variables is called a
2. The variable on the left-hand side is called the subject.

Using Formulas to Evaluate the Subject

We use formulas to find the value of the subject when we know the other values.

1. Write the formula.
2. Substitute given values.
3. Solve to find the unknown.

Example Substitute values into a formula and evaluate the subject

$$v = u + at$$

Find v when $u = 50, a = -10, t = 6$

1. $v = u + at$
2. $= 50 + (-10)(6)$
3. $= -10$

$$E = \frac{1}{2}mv^2$$

Find E when $m = 6, v = 10$

1. $E = \frac{1}{2}mv^2$
2. $= \frac{1}{2}(6)(10)^2$
3. $= 300$

Guided Practice

a $V = IR$

Find V when $I = 10, R = 250$

b $a = \frac{F}{m}$

Find a when $F = -80, m = 5$

c $C = I + 20x$

Find C when $I = -13, x = 15$

d $A = P(1 + r)^n$

Find A when $P = 1000, r = 0.05, n = 6$

e $c = \sqrt{a^2 + b^2}$

Find c when $a = 3, b = 4$

f $T = a + (n - 1)d$

Find T when $a = -3, n = 20, d = 2$

1 Consider the formula $A = 4p + 7$

a Find A if $p = 3$

$$A = 19$$

b Find A if $p = 11$

$$A = 51$$

c Find A if $p = -2$

$$A = -1$$

d Find A if $p = 13.5$

$$A = 61$$

2 Given these formulae, substitute known values and evaluate the subject.

a $V = IR$

find V when $I = 50$ and $R = 20$.

$$V = 1000$$

b $W = Fs$

find W when $F = 100$ and $s = 12$

$$W = 1200$$

c $s = \frac{d}{t}$

find s when $d = 50$ and $t = 10$

$$s = 5$$

d $v = u + at$

find v when $u = 5$, $a = 10$ and $t = 7$

$$v = 75$$

e $s = ut + \frac{1}{2}at^2$

find s when $u = -3$, $a = 10$ and $t = 5$

$$s = 110$$

f $d = 180 - \frac{360}{n}$

Find d when $n = 12$

$$d = 150$$

g $a = \frac{2Rn}{n+1}$

find a when $R = 8$, and $n = 24$

$$a = 15.36$$

h $V = \frac{4}{3}\pi r^3$

Find V when $r = 8.5$. Round to 2 d.p.

$$V = 2572.44$$

i $A = P(1 + r)^n$

find A when $P = 1000$, $r = 0.1$, $n = 5$

$$A = 1610.51$$

j $A = P(1 - r)^n$

find A when $P = 1000$, $r = 0.1$, $n = 5$

$$A = 590.49$$

k $c = \sqrt{a^2 + b^2}$

find c when $a = 5$, $b = 12$

$$c = 13$$

l $a = \sqrt{c^2 - b^2}$

find a when $c = 20$, $b = 16$

$$a = 12$$

- 3 Given these formulae, substitute known values and evaluate the subject.

a $S = V - Dn$

find S when $V = 17.5$, $D = 5$ and $n = 3$.

$$S = 2.5$$

b $z = \frac{x-y}{s}$

find z when $x = 24$, $y = 4.5$ and $s = 4$

$$z = 4.875$$

- 4 The formula used to convert between degrees Fahrenheit to degrees Celsius is $C = \frac{5}{9}(F - 32)$.

A recipe says to bake a roast at $400^\circ F$. How much is this in degrees Celsius?

Round your answer to the nearest degree.

$$C = 204^\circ C$$

- 5 The area of an annulus (or donut) can be calculated using $A = \pi(R^2 - r^2)$, where R is the radius of the large circle and r is the radius of the smaller circle.

Find the area of the annulus if $R = 8$ cm and $r = 4$ cm. Round your answer to 2 d.p.

$$A = \pi(64 - 16) \approx 150.80 \text{ cm}^2$$

- 6 Body mass index (BMI) is used to determine if an adult's mass is within a recommended range.

The formula for BMI is $b = \frac{m}{h^2}$ where m is mass in kilograms and h is height in metres.

- a** Mr Ding has a mass of 68 kg and height of 181 cm. Calculate his BMI.

$$b = 20.76$$

- b** Someone found Mr Ding's BMI to be 0.0020756... What mistake have they made?

They used height in centimetres instead of metres

- c** The recommended range for BMI is $21 < b < 25$. Based on Mr Ding's BMI, what would you recommend he do to get into the recommended range?

His BMI is slightly below 21, so he should gain a small amount of weight

7 2008 HSC Standard 2 Band 4

What is the value of $\sqrt{\frac{x+2y}{8y}}$ if $x = 5.6$ and $y = 3.1$, correct to 2 decimal places?

- (A) 0.69
- (B) 2.62
- (C) 2.83
- (D) 4.77

A (0.69)

8 HSC Standard 2 Sample Question Band 4

What is the value of $\frac{a+b}{ab}$ if $a = -2.1$ and $b = -3.6$, correct to 1 decimal place?

-0.8

9 HSC Standard 2 Sample Question Band 4

For adults (18 years and older), the Body Mass Index is given by $B = \frac{m}{h^2}$ where m is mass in kg and h is height in metres.

The medically accepted healthy range is $21 \leq B \leq 25$

What is the minimum weight for a 163 cm adult to be considered healthy?

Minimum healthy BMI is 21

$$21 = \frac{m}{1.63^2}$$

$$m = 21(1.63^2)$$

$$m = 55.79 \text{ kg}$$

Solving for an Unknown after Substitution

Review

☐ Solve equations

a $6y - 7 = 41$

b $4x - 3 = 17$

c $3g + 12 = 33$

d $10(q - 3) = 80$

e $\frac{m}{4} - 3 = 12$

f $\frac{n+3}{4} = 12$

☐ Substitute into formulas

a Substitute $y = 2$ into
 $y = 4x + 3$

b Substitute $A = 10, x = 3$ into
 $A = xy$

c Substitute $v = 5, a = -10, t = 3$ into
 $v = u + at$

d Substitute $c = 5, a = 3$ into
 $c^2 = a^2 + b^2$

Using Formulas to Find Unknowns

We use formulas to find the value of any unknown, if we know all other quantities.

1. Write the formula.
2. Substitute given values.
3. Solve to find the unknown.

Example Solve equations after substituting into a formula

$$y = 2x + 7.$$

Find x when $y = 29$.

1. $y = 2x + 7$
2. $29 = 2x + 7$
3. $22 = 2x$
 $11 = x$

$$A = (n - 1)d$$

Find n when $A = 40, d = 5$

1. $A = (n - 1)d$
2. $40 = (n - 1)(5)$
3. $8 = n - 1$
 $9 = n$

Guided Practice

a $A = lw.$

Find l if $A = 720$ and $w = 8$

b $y = 2x - 5$

Find x when $y = 20$

c $T = a(n + 1)$

Find n if $T = 55, a = 11$

d $A = P(1 + r)^n$

Find P if $A = 196, r = 0.4, n = 2$

Key Ideas

1. We can find any unknown value in a formula if we know the value of other variables.
 - a. Write the
 - b. given values
 - c. the equation using inverse operations

1 Consider the formula $U = 8a + 4$

a Find U if $a = -1$

$$U = -4$$

b Find a if $U = 44$. Set up an equation.

$$a = 5$$

c Find a if $U = 92$

$$a = 11$$

d If $U = -12$, find the value of a

$$a = -2$$

2 Consider the formula $y = 2x + 4$

a Find the value of x that makes $y = 16$

$$x = 6$$

b Find the value of x if $y = 0$

$$x = -2$$

3 Consider the formula $v = u + at$

a Find t when $v = 102$, $u = 18$ and $a = 7$

$$t = 12$$

b Find a when $v = 54$, $u = 12$ and $t = 14$

$$a = 3$$

4 Consider the formula $d = \frac{1}{2}ct$

a Find c when $d = 100$, $t = 8$

$$c = 25$$

b Find t when $d = 320$, $c = 18$

$$t = \frac{320}{9}$$

5 Consider the formula $I = Prn$

a Find P when $I = 19500$, $r = 0.03$, $n = 25$

$$P = 26000$$

b Find n when $I = 2100$, $P = 5000$, $r = \frac{6}{100}$

$$n = 7$$

- 6** A formula for the area of a rectangle is $A = bh$, where b is the base and h is the height.
- a** Calculate the area of a rectangle with base length of 8 cm and height of 9 cm.
- b** Calculate the base length of a rectangle with area of 20 cm^2 and height of 4 cm.

$$A = 72 \text{ cm}^2$$

$$b = 5 \text{ cm}$$

- c** Calculate the height of a rectangle with area of 25 cm^2 and base length of 5 cm.

$$h = 5 \text{ cm}$$

- 7** The perimeter of a rectangle of length l and width w is given by the formula $P = 2(l + w)$. Use the formula to:

- a** find P when $l = 6$ and $w = 20$

- b** find l when $w = 6$ and $P = 20$

$$P = 52$$

$$l = 4$$

- c** find w when $P = 10$ and $l = 2$

- d** find area of a rectangle if its perimeter is 20 and width is 4.

$$w = 3$$

$$l = 6, \text{ so } A = 24$$

- 8** The area of a triangle is given by the formula $A = \frac{1}{2}bh$. Use the formula to:

- a** find A when $b = 20$ and $h = 9$

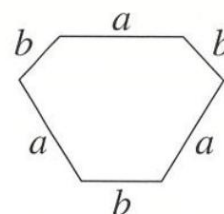
- b** find b when $A = 20$ and $h = 5$

$$A = 90$$

$$b = 8$$

- 9** Consider the diagram to the right.

- a** Find a formula for the perimeter, P , of the hexagon in terms of a and b



$$P = 3a + 3b$$

- b** Work out P when $a = 7$ and $b = 3$

- c** Work out a when $b = 2$ and $P = 21$

$$P = 30$$

$$a = 5$$

10 The side length of a square is l

a Write a formula to find the perimeter, P , of the square.

$$P = 4l$$

b Use the formula to find P when $l = 23$

$$P = 92$$

c Use the formula to form and solve an equation to find l when $P = 23$

$$l = 5.75$$

11 A rectangle has length l and width w . The formula for the area is $A = lw$

a Use the formula to find A when $l = 40$ and $w = 17$

$$A = 680$$

b The area of a different rectangle is 720 cm^2 . The length of this rectangle is 45 cm .
Use $A = lw$ to form an equation in w

$$720 = 45w$$

c Solve your equation to find the width of this rectangle.

$$w = 16 \text{ cm}$$

12 The cost $\$C$ to hire a taxi for a trip of length $d \text{ km}$ is $C = 3 + 2d$

a Find the cost of a 10 km trip. ($d = 10$)

$$c = \$23$$

b A trip cost a total of $\$161$. Determine the distance the taxi travelled.

$$79 \text{ km}$$

13 The volume of water (V litres) in a tank is given by $V = 4000 - 0.1t$,
where t is the time in seconds after a tap is turned on.

a Find the volume after 2 minutes .

$$3988 \text{ L}$$

b How long, to the nearest minute, does it take to empty the tank?

$$40\,000 \text{ seconds} \div 60 \approx 667 \text{ minutes}$$

14 A car salesman has a basic salary of \$800 a month.

For every car he sells, he earns a commission of \$150.

a Write a formula to find his wage, W , in terms of the number, n , of cars he sells.

$$W = 800 + 150n$$

b Use the formula to find his wage in a month when he sells 12 cars.

$$W = \$2600$$

c Use the formula to form and solve an equation to find the number of cars he sells in a month when he earns \$1850.

$$n = 7$$

15 Solve the equations to determine the value of the unknown pronumeral.

Round to 2 d.p. where appropriate.

a $F = ma$
 $F = 12, m = 3$

$$a = 4$$

b $m = \frac{F}{a}$
 $m = 12, a = 3$

$$F = 36$$

c $A = lb$
 $A = 30, l = 5$

$$b = 6$$

d $M = \frac{5k}{18},$
 $M = 15$

$$k = 54$$

e $C = \frac{890-2r}{n}$
 $C = 50, n = 10$

$$r = 195$$

f $v = u + at$
 $v = 6, u = 2, t = 4$

$$a = 1$$

g $S = \frac{n}{2}(a + l)$
 $S = 560, n = 20, l = 53$

$$a = 3$$

h $V^2 = gR,$
 $V = 12, g = 10$

$$R = 14.4$$

i $A = P \left(1 + \frac{r}{100} \right)$
 $A = 1000, P = 800$

$$r = 25$$

j $S = 2\pi r^2$
 $S = 72, \pi = 3.14$

$$r = 3.39$$

k $v^2 = u^2 + 2as$
 $v = 22, u = 6, a = 12$

$$s = 18.67$$

l $m = \frac{x}{\sqrt{y}}$
 $m = 8, y = 4$

$$x = 16$$

- 16** The formula for the area of a trapezium is $A = \frac{1}{2}h(a + b)$
- a** Find the value of A given that $a = 7$, $b = 8$ and $h = 10$

$$A = 75$$

- b** Find the value of h given that $A = 72$, $a = 6$ and $b = 10$

$$h = 9$$

- c** Find the value of b given that $A = 78$, $a = 7$ and $h = 12$

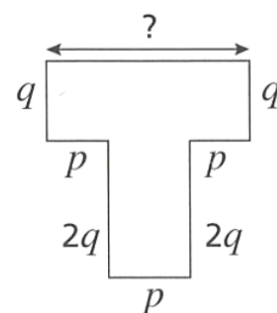
$$b = 6$$

- 17** Consider the diagram to the right.

- a** Explain why the length of the line with a question mark is $3p$.

Total width = $p + p + p$ from the stepped sides

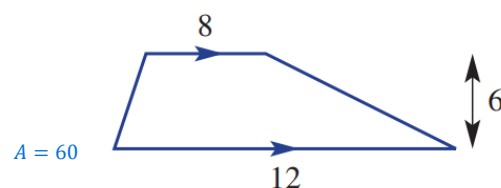
- b** Find a formula for the perimeter, P , of the shape,



$$P = 6p + 6q$$

- 18** The formula for the area of the trapezium is $A = \frac{1}{2}h(a + b)$, where a, b are the parallel sides and h is the perpendicular height.

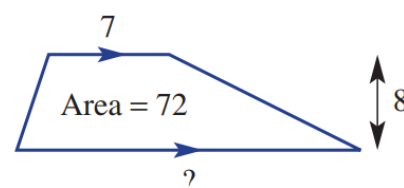
- a** Find the area of the trapezium on the right.



- b** Find the value of h if $A = 20$, $a = 3$ and $b = 7$

$$h = 4$$

- c** Find the missing value in the trapezium shown on the right.

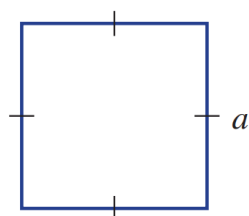


$$72 = 4(7 + b)$$

$$b = 11$$

19 Write a formula describing these relationships.

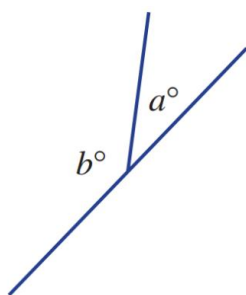
a



Perimeter = P

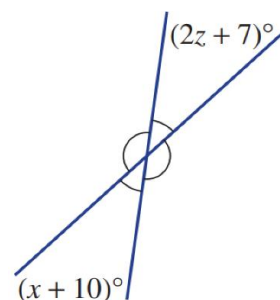
$$P = 4a$$

b



$$a + b = 180$$

c



$$2z + 7 = x + 10$$

20 $s = \frac{d}{t}$ gives the speed in km/h of a car that has travelled a distance of d km in t hours.
Calculate the time taken, to the nearest minute, to travel 380 km if the car averages 75 km/h.

$$t = \frac{380}{75} = 5.07$$

$$h \approx 5 \text{ h } 4 \text{ min}$$

21 To convert between temperatures in Celsius and Fahrenheit, use this formula:

$$F = \frac{9C}{5} + 32$$

a Find F if $C = 20$.

$$F = 68$$

b Find the value of C if $F = 50$.

$$C = 10$$

c When watching American TV, you hear "it's so hot, it's 100 degrees outside!" Fahrenheit is used in the USA. What is this in Celsius?

$$37.8^{\circ}\text{C}$$

22 Write a formula to show these mathematical relationships.
Make the first pronumeral the subject.

a How many $\$D$ you have, if you have c cents.

$$D = \frac{c}{100}$$

b How long something is in d cm, if it is m metres.

$$d = 100m$$

c The discounted price $\$D$ that is 30% off the original price $\$P$

$$D = 0.7P$$

d The cost $\$C$ of hiring a car at $\$50$ upfront plus $\$18$ per hour for t hours.

$$C = 50 + 18t$$

23 2015 HSC Standard 2 Band 4

The formula $C = \frac{5}{9}(F - 32)$ is used to convert temperatures between degree Fahrenheit (F) to degrees Celsius (C). Covert 3 degrees Celsius to the equivalent temperature in Fahrenheit.

$$\begin{aligned} 3 &= \frac{5}{9}(F - 32) \\ \frac{3 \times 9}{5} &= F - 32 \\ F &= \frac{27}{5} + 32 \\ &= 37.4 \end{aligned}$$

37.4°F

24 2019 HSC Standard 2 Band 4

Given the formula $C = \frac{A(y+1)}{24}$, calculate the value of y when $C = 120$ and $A = 500$

$$\begin{aligned} 120 &= \frac{500(y+1)}{24} \\ \frac{120 \times 24}{500} &= y + 1 \\ y &= \frac{120 \times 24}{500} - 1 \\ y &= 4.76 \end{aligned}$$

25 2020 HSC Standard 2 Band 4

The distance, d metres, travelled by a car slowing down from u km/h to v km/h can be obtained using the formula $v^2 = u^2 - 100d$.

What distance does as car travel while slowing down from 70 km/h to 40 km/h?

$$\begin{aligned} 40^2 &= 70^2 - 100d \\ 40^2 - 70^2 &= -100d \\ d &= \frac{40^2 - 70^2}{-100} \\ d &= 33 \text{ m} \end{aligned}$$

26 The velocity of an object is described by the formula $v = u + at$ where u is the initial velocity in m/s, a is acceleration in m/s², and t is the time in seconds.

Find the acceleration of a car that can go from 0 to 100 km/h in 7 seconds.

You can convert between speed in m/s and km/h using the formula $s_{\text{kmh}} = \frac{s_{\text{ms}}}{3.6}$.

$$\begin{aligned} 100 \text{ km/h} &= 27.78 \text{ m/s} \\ 27.78 &= 0 + 7a \\ a &\approx 3.97 \text{ m/s}^2 \end{aligned}$$

Challenge Exercise

- 1 Five more than twice a certain number is 33.

$$2x + 5 = 33$$

$$x = 14$$

- 2 Ten less than twice a number is equal to three more than the number.

$$2x - 10 = x + 3$$

$$x = 13$$

- 3 When three times a certain number is decreased by 4, it equals the number increased by 10.

$$3x - 4 = x + 10$$

$$2x = 14$$

$$x = 7$$

- 4 If a number is doubled and then increased by 3, it will be 8 more than its original value.

$$2x + 3 = x + 8$$

$$x = 5$$

- 5 When a number is added to a third of the number, it equals 20.

$$x + \frac{x}{3} = 20$$

$$3x + x = 60$$

$$4x = 60$$

$$x = 15$$

- 6 When a number is decreased by a quarter of the number, it equals 12.

$$x - \frac{x}{4} = 12$$

$$4x - x = 48$$

$$3x = 48$$

$$x = 16$$

- 7 If a number is halved and then increased by 10, it will be 4 more than its original value.

$$\frac{x}{2} + 10 = x + 4$$

multiply by 2

$$x + 20 = 2x + 8$$

$$x = 12$$

- 8 If 25 is added to one-third of a number, the result is double the number.

$$\frac{x}{3} + 25 = 2x$$

multiply by 3

$$x + 75 = 6x$$

$$5x = 75$$

$$x = 15$$

- 9** Adding 3 to half a certain number gives the same result as subtracting 9 from twice the number.

$$3 + \frac{x}{2} = 2x - 9$$

multiply by 2

$$6 + x = 4x - 18$$

$$3x = 24$$

$$x = 8$$

- 10** Find the numbers in the following:

- a** The sum of two consecutive odd numbers is 32.

Let smaller odd number be x .

$$x + (x + 2) = 32,$$

$$x = 15.$$

Numbers are 15 and 17.

- b** The sum of four consecutive numbers is 138.

Let numbers be $x - 1$, x , $x + 1$, $x + 2$.

$$4x + 2 = 138, x = 34.$$

Numbers are 33, 34, 35 and 36.

- c** The sum of two numbers is 30, and the second number is four times the first.

Let first number be x , second is $4x$.

$$5x = 30, x = 6.$$

Numbers are 6 and 24.

- d** The sum of three consecutive odd numbers is 23 more than the sum of the even numbers that lie between them.

Let odd numbers be $2x + 1$, $2x + 3$, $2x + 5$; even numbers between them:

$$2x + 2, 2x + 4.$$

$$(2x + 1) + (2x + 3) + (2x + 5) = (2x + 2) + (2x + 4) + 23$$

$$6x + 9 = 4x + 29,$$

$$2x = 20, x = 10.$$

Numbers are 21, 23 and 25.

- 11** If a drink and a sandwich together cost \$5, and the sandwich costs 60¢ more than the drink, how much does the drink cost?

Let drink cost x cents, sandwich costs $x + 60$ cents.

$$x + (x + 60) = 500,$$

$$2x = 440,$$

$$x = 220.$$

The drink costs \$2.20.

- 12** A book and a card cost \$28.
If the book costs 6 times as much as the card, find the price of each.

Let card cost \$ c , book costs \$ $6c$.

$$6c + c = 28,$$

$$7c = 28,$$

$$c = 4.$$

Card: \$4, book: \$24.

- 13** Linda is five times as old as her daughter Mia.
Together their ages add up to 36 years. How old are they?

Let Mia's age be m , Linda's age is $5m$.

$$m + 5m = 36, m = 6.$$

Mia is 6, Linda is 30.

- 14** A pair of shoes and a matching handbag cost \$180. If the shoes cost \$25 more than the bag, find the cost of each.

Let bag cost \$ y , shoes cost \$ $(y + 25)$. $y + (y + 25) = 180$,

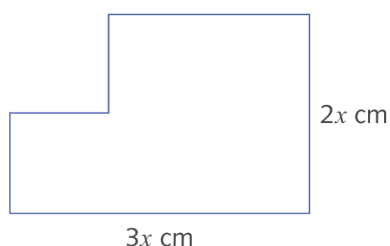
$$2y = 155,$$

$$y = 77.50.$$

Bag: \$77.50, shoes: \$102.50.

15

- a** Find the perimeter of the figure shown below in terms of x .



total horizontal extent = $2 \times 3x$ and total vertical extent = $2 \times 2x$.

Perimeter = $10x$ cm.

- b** If the perimeter is 64 cm, find the value of x .

$$10x = 64,$$

$$x = 6\frac{2}{5}$$

- 16** In a maths lesson Jamie drew some triangles and some quadrilaterals.
He drew 27 polygons altogether and a total of 99 vertices. How many triangles did he draw?

Let x = triangles, $27 - x$ = quadrilaterals.

$$3x + 4(27 - x) = 99$$

$$-x = -9,$$

$$x = 9.$$

Jamie drew 9 triangles.

- 17** In the park, Doyen saw people and dogs. Altogether, he counted 10 heads and 26 legs.
How many dogs and how many people did he see?

Let x = people, $10 - x$ = dogs.

$$2x + 4(10 - x) = 26$$

$$-2x = -14, x = 7.$$

7 people and 3 dogs.

- 18** In a sport store golf balls are sold either boxed in threes or fours.
If there are five times as many boxes of threes as there are boxes of fours, and there are 361 golf balls altogether, find how many boxes have four golf balls in them.

Let n = boxes of 4, so $5n$ = boxes of 3.

$$4n + 3(5n) = 361$$

$$19n = 361, n = 19.$$

19 boxes of four golf balls.

- 19** A retailer buys tennis rackets at \$84 each and squash rackets at \$54 each.
He bought 40 rackets altogether, and the total cost was \$2520. How many of each did he buy?

Let r = number of tennis rackets, $40 - r$ = squash rackets.

$$84r + 54(40 - r) = 2520$$

$$30r = 360,$$

$$r = 12.$$

12 tennis rackets and 28 squash rackets.

- 20** A tape and a CD costs \$24, while two tapes and three CDs cost \$63.
How much does each cost?

Let tape cost $\$t$, CD costs $\$(24 - t)$.

$$2t + 3(24 - t) = 63$$

$$72 - t = 63,$$

$$t = 9.$$

Tape: \$9, CD: \$15.

- 21** Let n be the age of a student (in years).
The student's sister is 3 years older than the student, and the student's father is 25 years older than the student.
The sum of the ages of the student and his sister is 11 less than the age of their father.
Find the ages of the student, his sister and their father.

Student: n ; sister: $n + 3$; father: $n + 25$.

$$n + (n + 3) = (n + 25) - 11$$

$$2n + 3 = n + 14,$$

$$n = 11.$$

Student: 11, sister: 14, father: 36.

- 22** Julian is 18 years old, while his mother is 46.
In how many years time will his mother be twice as old as Julian?

Let this happen in x years. $46 + x = 2(18 + x)$,

$$x = 10.$$

In 10 years.

- 23** David is 30 years older than his son Edward.
In 6 years time he will be 4 times Edward's age. How old is Edward currently?

Let Edward's age be x , David's is $x + 30$.

$$(x + 30) + 6 = 4(x + 6)$$

$$x + 36 = 4x + 24,$$

$$3x = 12,$$

$$x = 4.$$

Edward is 4.

- 24** Four years ago, I was twice as old as I was 14 years ago. What is my age?

Let my age be x .

$$x - 4 = 2(x - 14),$$

$$x = 24. \text{ I am 24 years old.}$$

- 25** Natalie is 5 times as old as her son Jeremy.
In 8 years time, she will be 3 times as old as her son. Find Natalie's age now.

Let Jeremy's age be x , Natalie's is $5x$.

$$5x + 8 = 3(x + 8)$$

$$2x = 16,$$

$$x = 8.$$

Natalie is 40.

- 26** If Jeremy is now three times as old as Brian, how many years ago was Brian 3 and Jeremy 35?

Let it be x years ago; Brian was 3, Jeremy was 35.

Currently Jeremy = $35 + x$, Brian = $3 + x$, and Jeremy is now three times Brian:

$$35 + x = 3(3 + x) \quad 2x = 26, \quad x = 13.$$

13 years ago.

- 27** Jasmine, Frank and Richard all have their birthdays on the same day.
Richard's present age is two years less than the sum of Jasmine's and Frank's present ages.
Frank is twice as old as Jasmine.
In 6 years time Richard will be twice as old as Jasmine.
At present, the sum of all their ages is 46. How old is each person now?

Let Jasmine's age be x ; Frank: $2x$; Richard: $(x + 2x) - 2 = 3x - 2$.

$$x + 2x + (3x - 2) = 46$$

$$6x = 48,$$

$$x = 8.$$

Jasmine: 8, Frank: 16, Richard: 22.

- 28** Two brothers are given the same amount of money for Christmas. Initially Ben had \$143 and Michael \$19, and now Ben has five times as much as Michael. How much are the brothers given for Christmas?

Let amount given be $\$x$.

$$143 + x = 5(19 + x)$$

$$4x = 48,$$

$$x = 12.$$

Each brother was given \$12.

- 29** Renae and James have \$60 between them. Renae gives James \$12. She now has twice as much as James. How much did she have previously?

Let Renae have $\$r$. After transfer: Renae has $r - 12$, James has $(60 - r) + 12 = 72 - r$.

$$r - 12 = 2(72 - r)$$

$$3r = 156,$$

$$r = 52.$$

Renae had \$52.

- 30** Morrie spent \$9 on pencils and rulers for his class. How many of each did he buy if:

- a** pencils cost 20 cents each and rulers 50 cents each, and he bought twice as many pencils as rulers?

Let x = number of rulers, $2x$ = pencils (in cents):

$$50x + 20(2x) = 900$$

$$90x = 900,$$

$$x = 10.$$

10 rulers and 20 pencils.

- b** pencils cost 50 cents and rulers cost 80 cents each?

Let p = pencils, r = rulers. $50p + 80r = 900$,

$$5p + 8r = 90.$$

Integer solutions: $(p, r) = (10, 5)$ or $(p, r) = (2, 10)$.

- 31** A certain examination has ten problems, each requiring a single answer which must be either right or wrong. It is marked by giving 2 for each correct answer and -1 for each incorrect answer. Give a formula for the marks (m) obtained by a student who answers all ten parts, but only gets c parts correct.

c correct,

$10 - c$ incorrect.

$$m = 2c - (10 - c) = 3c - 10$$

- 32** In a multiple-choice exam of 30 questions, a correct answer earns 2 marks, an incorrect answer loses 3 and an unanswered question scores 0 marks.

If Glen left out 6 questions and his score was 28, how many questions did he get correct?

Glen answered $30 - 6 = 24$ questions.

Let c = correct, $24 - c$ = incorrect.

$$2c - 3(24 - c) = 28$$

$$5c = 100,$$

$$c = 20.$$

Glen answered 20 correctly.

- 33** Find out how many marbles each person has.

- a** Graham has 3 less marbles than Hazel, and Joseph has twice as many marbles as Graham. Altogether they have 119 marbles.

Let Hazel have x ; Graham: $x - 3$; Joseph: $2(x - 3)$.

$$x + (x - 3) + 2(x - 3) = 119$$

$$4x = 128,$$

$$x = 32.$$

Hazel: 32, Graham: 29, Joseph: 58.

- b** Evie has 3 times as many marbles as John, but 9 less marbles than Peter. Altogether they have 93 marbles.

Let John have x ; Evie: $3x$; Peter: $3x + 9$.

$$x + 3x + (3x + 9) = 93$$

$$7x = 84,$$

$$x = 12.$$

John: 12, Evie: 36, Peter: 45.

- c** Ilana has 5 more marbles than Doug. Judy has twice as many as Ilana and Doug combined. Altogether they have 135 marbles.

Let Doug have x ; Ilana: $x + 5$; Judy: $2(x + x + 5) = 4x + 10$.

$$x + (x + 5) + (4x + 10) = 135$$

$$6x = 120, x = 20.$$

Doug: 20, Ilana: 25, Judy: 90.

- d** Kate has twice as many marbles as Don.
If Kate gave Don 5 marbles, she will still have 10 more marbles than Don.

Let Don have x , Kate has $2x$.

After transfer: Kate has $2x - 5$, Don has $x + 5$.

$$2x - 5 = (x + 5) + 10,$$

$$x = 20.$$

Don: 20, Kate: 40.

- 34** In another examination with 30 problems, each correct question is worth 5 marks; if a question is not attempted, no marks are deducted, but if it is incorrect 2 marks are deducted. If Jennifer attempts 26 questions and gets c of these correct, find a formula for the number of marks she is awarded.

c correct, $26 - c$ incorrect, 4 unanswered.

$$m = 5c - 2(26 - c) = 7c - 52$$

- 35** Choose any three-digit number. Reverse the order of the digits. Now subtract the smaller number from the larger. Show that your result is always divisible by 99 by trying several examples. Justify that this will always be the case algebraically by letting the original three-digit number be $100a + 10b + c$.

Reversed number: $100c + 10b + a$.

$$\begin{aligned}\text{Difference (larger minus smaller): } (100a + 10b + c) - (100c + 10b + a) &= 99a - 99c \\ &= 99(a - c).\end{aligned}$$

Always divisible by 99.

- 36** I have a Magic Money Box which will double any amount of money placed in it, then add \$1 to the amount. One day, I placed a certain amount of money in the Magic Money Box and got a new amount. I then placed the new amount in the Magic Money Box and got \$75. How much money did I first place in the box?

Let original amount be $\$x$.

After first use: $2x + 1$.

After second use: $2(2x + 1) + 1 = 75$.

$$4x + 3 = 75$$

$$4x = 72,$$

$$x = 18.$$

\$18 was placed in the Magic Money Box.

- 37** Write down any four-digit number, reverse the digits, subtract the smaller number from the larger number and add the digits of your answer. Repeat for several numbers. Write down your findings and prove it using algebra.

Let the four-digit number be $1000a + 100b + 10c + d$.

Reversed: $1000d + 100c + 10b + a$.

$$\text{Difference: } 999a + 90b - 90c - 999d = 9(111a + 10b - 10c - 111d).$$

The result is always divisible by 9, so the digit sum of the answer is always a multiple of 9.

38 Think of a number. Let this number be x .

a Write the following using algebra to see what you get:

Add 6 to the number, then

multiply the result by 10, then

subtract the result from 20, then

multiply this result by 2.

Step by step: $x + 6$,

$$10(x + 6),$$

$$20 - 10(x + 6),$$

$$2(20 - 10(x + 6))$$

$$= -80 - 20x$$

b If this final result is twice the original number, find the original number x .

$$-80 - 20x = 2x,$$

$$-80 = 22x,$$

$$x = -3\frac{7}{11}$$

c If the number obtained in part (a) is 1000 less than the original number, find the original number x .

$$-80 - 20x = x - 1000,$$

$$21x = 920,$$

$$x = 43\frac{17}{21}$$

39 Humphrey leaves Tantown at 2 p.m. and travels at 60 km/h along the Trans Highway. He passes through Sintown, which is 16 km from Tantown along the Trans Highway. Petro leaves Sintown at 3 p.m. and travels at 80 km/h along this highway in the same direction that Humphrey is travelling in. When and where does Petro catch up to Humphrey?

At 3 p.m., Humphrey is 60 km from Tantown; Petro is 16 km from Tantown.

Let t = hours after 3 p.m.

Humphrey's distance from Tantown: $60 + 60t$

Petro's distance from Tantown: $16 + 80t$

Setting equal: $60 + 60t = 16 + 80t$

$$20t = 44,$$

$$t = 2.2 \text{ h (2 h 12 min).}$$

Petro catches Humphrey at 5:12 p.m.,

$$60 + 60(2.2) = 192 \text{ km from Tantown.}$$